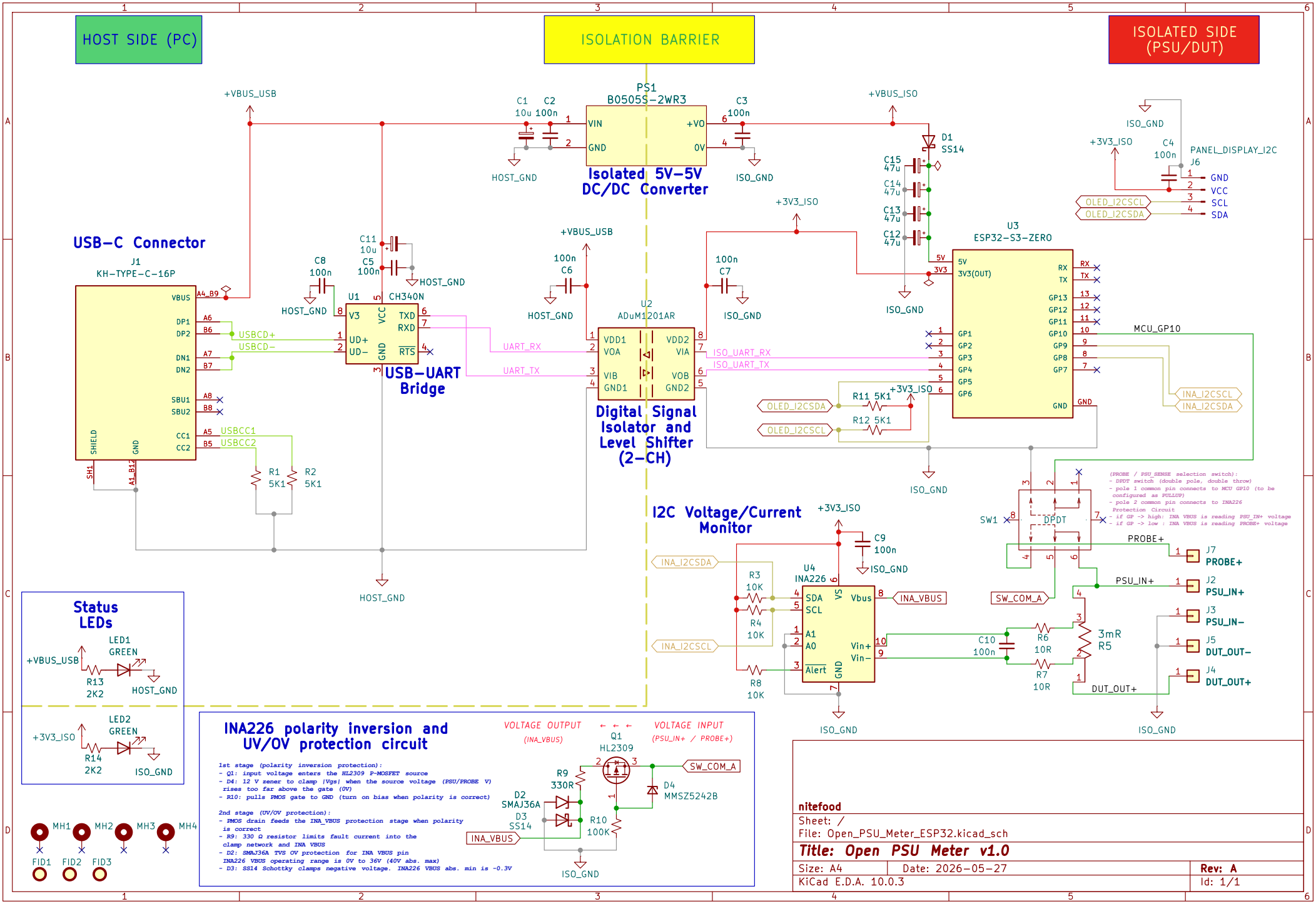
HOST SIDE (PC)
ISOLATION BARRIER
ISOLATED SIDE (PSU/DUT)
USB-C Connector
J1
KH-TYPE-C-16P
Isolated 5V-5V DC/DC Converter
PS1
B0505S-2WR3
USB-UART Bridge
U1
CH340N
Digital Signal Isolator and Level Shifter (2-CH)
U2
ADuM1201AR
U3
ESP32-S3-ZERO
I2C Voltage/Current Monitor
U4
INA226
Status LEDs
LED1 GREEN
LED2 GREEN
INA226 polarity inversion and UV/OV protection circuit
1st stage (polarity inversion protection):
- Q1: input voltage enters the HL2309 P-MOSFET source
- D4: 12 V zener to clamp |Vgs| when the source voltage (PSU/PROBE V) rises too far above the gate (0V)
- R10: pulls PMOS gate to GND (turn on bias when polarity is correct)
2nd stage (UV/OV protection):
- PMOS drain feeds the INA_VBUS protection stage when polarity is correct
- R9: 330 Ω resistor limits fault current into the clamp network and INA VBUS
- D2: SMAJ36A TVS OV protection for INA VBUS pin
INA226 VBUS operating range is 0V to 36V (40V abs. max)
- D3: SS14 Schottky clamps negative voltage. INA226 VBUS abs. min is -0.3V
VOLTAGE OUTPUT (INA_VBUS)
VOLTAGE INPUT (PSU_IN+ / PROBE+)
Q1 HL2309
D2 SMAJ36A
D3 SS14
D4 MMSZ5242B
R9 330R
R10 100K
(PROBE / PSU_SENSE selection switch):
- DPDT switch (double pole, double throw)
- pole 1 common pin connects to MCU GP10 (to be configured as PULLUP)
- pole 2 common pin connects to INA226 Protection Circuit
- if GP -> high: INA VBUS is reading PSU_IN+ voltage
- if GP -> low : INA VBUS is reading PROBE+ voltage
nitefood
Sheet: /
File: Open_PSU_Meter_ESP32.kicad_sch
Title: Open PSU Meter v1.0
Size: A4
Date: 2026-05-27
Rev: A
KiCad E.D.A. 10.0.3
Id: 1/1